

Distributionally Robust Optimization for Aerial Multi-Access Edge Computing via Cooperation of UAVs and HAPs

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Abstract—With an extensive increment of computation demands, the aerial multi-access edge computing (MEC), mainly based on uncrewed aerial vehicles (UAVs) and high altitude platforms (HAPs), plays significant roles in future network scenarios. In detail, UAVs can be flexibly deployed, while HAPs are characterized with large capacity and stability. Hence, in this paper, we provide a hierarchical model composed of an HAP and multi-UAVs, to provide aerial MEC services. Moreover, considering the errors of channel state information from unpredictable environmental conditions, we formulate the problem to minimize the total energy cost with the chance constraint, which is a mixed-integer nonlinear problem with uncertain parameters and intractable to solve. To tackle this issue, we optimize the UAV deployment via the weighted K-means algorithm. Then, the chance constraint is reformulated via the distributionally robust optimization (DRO). Furthermore, based on the conditional value-at-risk mechanism, we transform the DRO problem into a mixed-integer second order cone programming, which is further decomposed into two subproblems via the primal decomposition. Moreover, to alleviate the complexity

of the binary subproblem, we design a binary whale optimization algorithm. Finally, we conduct extensive simulations to verify the effectiveness and robustness of the proposed schemes by comparing with baseline mechanisms.

Index Terms—Aerial multi-access edge computing, resource allocation, distributionally robust optimization (DRO), conditional value-at-risk (CVaR), primal decomposition, binary whale optimization (BWOA).

I. INTRODUCTION

IN RECENT years, with the extensive growth of computational intensive tasks, the multi-access edge computing (MEC) technique is raised to provide services for various ground users (GUs) in the sixth generation (6G) communication networks [1], [2], [3], [4]. However, as for the GUs in remote areas such as deserts and oceans, it lacks ground infrastructures to provide communication coverage and MEC services [5], [6]. As forecasted that the global uncrewed aerial vehicles (UAVs) market is expected to reach the worth of \$55.8 billion by 2030, UAVs show the colossal potential in the applications of future industries [7]. The non-terrestrial networks can provide ubiquitous coverage for the remote GUs, in which UAVs can be flexibly and quickly deployed on demand with low cost [8], [9], [10], [11], [12], [13]. However, UAVs are limited by the load capacity of computing module, battery, etc. Alternatively, the high altitude platform (HAP), suspending above 20 km at a quasi-static position, and equipped with stronger computing resources and sufficient energy, can compensate for the resource-limited UAVs [14], [15]. Besides, compared with HAPs, UAVs are relatively flexible and can be rapidly deployed to meet the sudden surge in data requests. Therefore, in the context of 6G networks, the cooperation of UAVs and HAPs can provide flexible and stable aerial MEC for GUs in diverse applications to reduce latency and improve the quality of service (QoS).

Unfortunately, in the aerial MEC network, the tasks generated from GUs may be heterogeneous with different QoS demands such as the tolerated delay. Besides, the resources of aerial MEC networks, such as communication, computing, and energy are limited, in which the energy supply is the basic for all operations [16]. Hence, how to guarantee the QoS of GUs and take full advantage of aerial resources is a key issue for ubiquitous communication and computation services

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in the 6G networks [17], [18]. Furthermore, considering the unpredictable environmental fluctuations, the communication link from the GU to UAV (G2U) is highly dynamic, and the channel state information (CSI) is imperfect, which may cause errors and mismatches between the realistic situation and ideal circumstance [19]. Such errors bring more challenges for the resource allocation scheme in the aerial MEC. Besides, how to cooperate UAVs and HAPs for efficient data offloading and resource optimization is also challenging.

To deal with the above challenges, in this paper, we propose an aerial MEC framework composed of UAVs and an HAP to cooperatively serve the GUs in remote areas, with the consideration of imperfect transmission CSI. In detail, an uncertainty set is constructed to capture the potential random parameters and a chance constraint for task latency with CSI estimation errors is formulated. Then, considering multi-resource constraints for UAVs and the HAP, we formulate the problem to minimize the total energy consumption, with regard to UAV positions, GU-UAV connection decisions, offloading strategies and resource allocation. Since the problem is in the form of mixed integer non-linear programming (MINLP) and NP-hard to solve [20], we first cluster the GUs and deploy UAVs at appropriate positions via proposing the weighted K-means deployment (WKD) based algorithm with a low time complexity. Taking into account the different characteristics of tasks, the weighted distance metric is applied so that the importance of different tasks is incorporated. Then, we reformulate the chance constraint without distribution information into a mixed integer second order cone programming (MISOCP) form by employing the distributionally robust optimization (DRO) and conditional value-at-risk (CVaR) mechanism. To reduce the complexity, via the primal decomposition, we further decompose the MISOCP problem into two subproblems with respect to the offloading decisions and computing resource allocation, respectively. The problem concerning resource allocation is solved by a standard convex toolkit. Moreover, to tackle the integer programming problem related to the offloading decision, we design a meta-heuristic algorithm termed as binary whale optimization algorithm (BWOA).

The main contributions of this work are summarized as follows.

- We propose a hierarchical aerial MEC model composed of an HAP and multi-UAVs to provide services for remote GUs, in which UAVs can be deployed flexibly and the HAP provides stable and strong computing services. Besides, the CSI estimation error is modeled by an uncertainty set based on the historical statistical information and the time latency requirement is formulated as a chance constraint.
- To handle the problem of multi-UAV deployment, a WKD based algorithm is designed. Then, by the DRO and CVaR based mechanism, the chance constraint is reformulated into an MISOCP form.
- To tackle the reformulated mixed integer programming (MIP) problem with the MISOCP constraint, by leveraging the primal decomposition, it is decomposed into two subproblems. The subproblem on the resource allocation is convex and solved via CVX. To further reduce the

complexity of the binary offloading subproblem, we design the BWOA.

- Extensive simulations are conducted to evaluate the proposed algorithms under various circumstances. The robustness of the designed algorithms with CSI estimation errors is verified. Moreover, by comparing with other baseline algorithms, the effectiveness and low-complexity of the proposed algorithms are verified.

The rest of this paper is arranged as follows. Related works are presented in Section II. Section III proposes the system model and problem formulation. Algorithms are designed in Section IV. Simulations and numerical results are provided in Section V. Finally, we draw conclusions in Section VI.

II. RELATED WORKS

As for the UAV-based MEC, there exist abundant recent researches. For instance, [21] proposed a collaborative MEC frame and exploited a deep reinforcement learning method to jointly optimize resource allocation, UAV trajectory, and task scheduling. [22] discussed a multi-UAV-enabled MEC system and solved the decoupled two subproblems via alternating optimization and successive convex approximation mechanisms. Considering the competitive relationship among UAVs, [23] formulated a joint optimization problem for the multi-dimensional resource constrained UAV-MEC network and put forward a triple learner based approach. While these studies made significant progresses in the resource and trajectory optimization, improved frameworks are still worth considering. In [24], the authors designed a three-stage alternating algorithm to address issues concerning energy consumption in the UAV-related MEC system. In [25], a deep reinforcement learning approach was devised to minimize the computation cost in the multi-UAV based MEC system. [26] investigated a multi-objective optimization problem in the MEC network to minimize the delay and energy consumption as well as maximize the number of collected tasks of UAVs. In [27], a hierarchical UAV-assisted MEC framework was studied to minimize the sum of latency and energy consumption, in which a method by jointly combining deep reinforcement learning and convex optimization was designed. [28] presented a two-layer optimization framework to reduce the energy consumption for the UAV-based MEC. The authors in [29] designed a two-layer optimization approach and tackled the 0-1 integer programming problem by a greedy algorithm, in which by jointly optimizing task scheduling and UAV locations, the energy consumption in the MEC system was minimized. [30] designed a robust multi-agent approximation strategy to address the uncertainties of CSI and task complexity, which was solved by the multi-agent deep reinforcement learning. However, limited by their battery capacities, the applications of UAVs are restricted in terms of the large-scale service provision for delay sensitivity and computation intensive tasks.

Although the above researches have made great progresses in the resource and trajectory optimization, UAVs are limited by their battery capacities and have shortcomings in providing better services for delay sensitive and computationally intensive tasks. Different from UAVs with constrained capabilities, HAPs

can provide strong payloads and stable coverage, which contributes to completing intensive MEC services. In recent years, some works begin to explore the applications of HAPs based MEC services. For example, [31] jointly deployed multi-UAVs and an HAP to provide connectivity and computing service for GUs. [32] focused on data offloading in the UAV-HAP MEC system and developed a matching-based algorithm to maximize the total processed data. In [33], a multi-dimensional resource allocation problem in the UAV-HAP system was designed to minimize the average age of information in response to the uncertain errors of CSI, and a learning-based algorithm was presented to tackle this non-convex problem. Due to the limited resources of the aerial MEC platforms, more studies focused on the issue to improve the energy consumption and resource utilization. In [34], a resource allocation problem minimizing the energy cost was studied for the UAV-HAP assisted MEC, and solved by the distributed online algorithm based on the game theory. The authors in [35] focused on the energy-efficient trajectory optimization problem in the UAV-HAP based MEC system and designed a modified multi-objective reinforcement learning algorithm. [36] employed the K-means and multi-agent reinforcement learning algorithms for resource utilization in the UAV-HAP assisted MEC system. The authors in [37] investigated a computation offloading problem in the UAV-HAP aerial MEC system, and a markov game was conducted to enhance the energy harvesting performance for UAVs. [38] built a multi-objective Markov decision process model towards the age of information and energy tradeoff problem in which UAVs and HAPs cooperatively provided MEC services for ground devices.

As analyzed above, the cooperation of UAVs and HAPs leverages the advantages of the aerial platforms for better energy efficiency, lower latency, and higher capability. Nevertheless, in these studies, the CSI errors are mostly ignored or following a specific distribution, which is impractical due to the various interferences caused by actual environment. The inability to estimate accurate CSI in the practical scenarios brings more challenges for efficient resource allocations. Therefore, it is essential to exploit robust algorithms to deal with the unpredictable fluctuation from the environment. Based on above considerations, in this paper, we focus on the cooperation of UAVs and the HAP to provide robust aerial MEC services for GUs, with the consideration of the imperfect CSI by the uncertainty set.

III. SYSTEM MODEL AND PROBLEM FORMULATION

In this section, a two-layer aerial MEC model is proposed in Section III-A. Then, the communication model and offloading model are proposed in Sections III-B and III-C, respectively. The problem formulation is detailed in Section III-D.

A. Aerial MEC Model

As shown in Fig. 1, a hierarchical aerial MEC system is proposed. M GUs indicated by the set $\mathcal{M} = \{1, 2, \dots, M\}$, $m \in \mathcal{M}$, are randomly distributed within the remote areas. N UAVs equipped with edge servers, the set of which is denoted by $\mathcal{N} = \{1, 2, \dots, N\}$, $n \in \mathcal{N}$, are deployed to provide MEC services. An HAP, denoted by h , hovers at a fixed position and acts

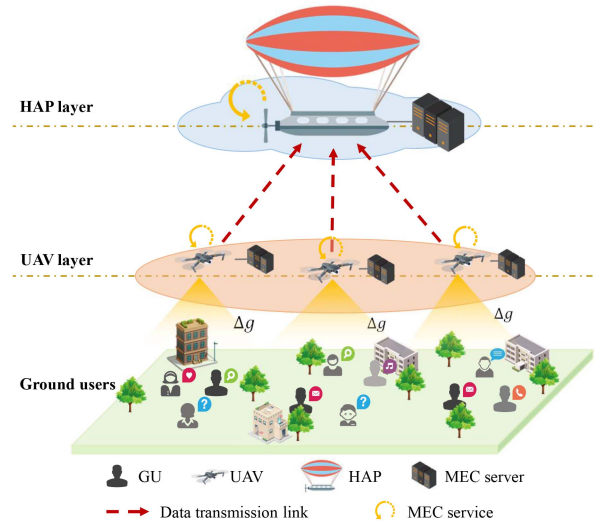


Fig. 1. Aerial MEC model composed of UAVs and an HAP.

as the supplement for the resource-limited UAVs. The Cartesian coordinate is utilized to represent the locations. $\mathbf{w}_m$ denotes the location of GU m , where $\mathbf{w}_m = (x_m, y_m)$. All UAVs are assumed to hover at the same altitude z_n , and the horizontal deployment position of UAV n is denoted by $v_n = (x_n, y_n)$. After deployment, the UAVs hover in the air and are regarded as quasi-stationary. The distance between GU m and UAV n is calculated as:

$$d_{m,n} = \sqrt{(x_m - x_n)^2 + (y_m - y_n)^2 + z_n^2}, \forall m \in \mathcal{M}, \forall n \in \mathcal{N}. \quad (1)$$

Since UAVs are equipped with limited computing resources, an HAP with strong capabilities is deployed in the upper layer to assist with computing, whose location is denoted by $\varpi_h = (x_h, y_h, z_h)$. Therefore, the distance between UAV n and HAP h is

$$d_{n,h} = \sqrt{(x_n - x_h)^2 + (y_n - y_h)^2 + (z_n - z_h)^2}, \forall n \in \mathcal{N}. \quad (2)$$

The task generated from GU m is denoted as $(L_m, c_m, T_m^{\max})$, where L_m represents the size of task data, c_m indicates the required CPU cycles to process 1 b data and $T_m^{\max}$ is the maximum tolerable delay of task m . To guarantee the delay limitation, the task needs to be completed within $T_m^{\max}$. We consider that the task cannot be divided and the task offloading pattern is the binary mode. Let binary variable δ_m^n indicate the connection relationship between GU m and UAV n , i.e.,

$$\delta_m^n = \begin{cases} 1, & \text{GU } m \text{ is connected with UAV } n, \\ 0, & \text{otherwise.} \end{cases} \quad (3)$$

Considering the accessing constraint, each GU can only connect to one UAV, we have

$$\sum_{n=1}^N \delta_m^n = 1, \forall m \in \mathcal{M}. \quad (4)$$

If a UAV is not able to provide sufficient computing resources for the task, or the delay limitation is unable to be satisfied, then

the task is forwarded to the HAP for processing. In this case, the UAV performs as a relay. In detail, binary variable λ_m^n is introduced to indicate whether the task collected by UAV n is forwarded to the HAP, i.e.,

$$\lambda_m^n = \begin{cases} 1, & \text{task } m \text{ is forwarded to the HAP by UAV } n, \\ 0, & \text{otherwise.} \end{cases} \quad (5)$$

The HAP is assumed to process at most H tasks simultaneously, and so we have

$$\sum_{m=1}^M \sum_{n=1}^N \lambda_m^n \leq H. \quad (6)$$

B. Communication Model

1) *G2U Channel Model*: The G2U channel is a large-scale fading model [39] and can be regarded as a line-of-sight channel. The uplink transmission channel gain under ideal condition of GU m is given as:

$$\bar{g}_m^u = \sum_{n=1}^N \frac{\delta_m^n g_0^u}{d_{m,n}^2}, \forall m \in \mathcal{M}, \quad (7)$$

where g_0^u is the power gain at the reference distance $d_0 = 1$ m.

Since the G2U channel is time-varying and vulnerable with the impacts from obstacles, complicated terrains and electromagnetic interferences, the CSI cannot be obtained precisely. In other words, there exist CSI estimation errors between the ideal and realistic environments, due to the inevitable disturbances and interferences from the environment. Accordingly, we denote the actual G2U channel gain as g_m^u :

$$g_m^u = \bar{g}_m^u + \Delta_m, \forall m \in \mathcal{M}, \quad (8)$$

where Δ_m is the unmeasurable CSI estimation error. Clearly, it is difficult to obtain accurate results or probability distributions of the CSI errors in real situations. Therefore, we consider that the moment estimation information can be obtained from the historical statistical data. In particular, the uncertainty set $\mathcal{P}$ is constructed to describe all the possible distributions of the random errors, i.e.,

$$\mathcal{P} = \left\{ \mathbb{P} \in \mathcal{P} \left| \begin{array}{l} \mathbb{E}_{\mathbb{P}}(\Delta_m) = \mu_m, \\ \mathbb{D}_{\mathbb{P}}(\Delta_m) = \sigma_m^2, \end{array} \right. \right\}, \quad (9)$$

where μ_m is the mean of random parameters Δ_m under distribution $\mathbb{P}$, and σ_m^2 is the corresponding variance. The uncertainty set $\mathcal{P}$ comprises all possible probability distributions of the random CSI estimation error Δ_m , i.e., $\mathbb{P} \in \mathcal{P}$. Besides, to simplify the communication model, we adopt the orthogonal frequency division multiple access (OFDMA) technology. In this way, GUs are enabled to transmit their data simultaneously, and the mutual interference is correspondingly ignored. According to the Shannon formula, the uplink rate of G2U channel is

$$r_m^u = B_u \log_2 \left(1 + \frac{p_u g_m^u}{n_0 B_u} \right), \forall m \in \mathcal{M}, \quad (10)$$

where B_u denotes the bandwidth allocated to each task, p_u is the transmitting power of GUs, and n_0 represents the power

spectrum density of additive white noise. Then, the uplink transmission delay of GU m is

$$t_m^u = \frac{L_m}{r_m^u}, \forall m \in \mathcal{M}. \quad (11)$$

2) *U2H Channel Model*: Different from the vulnerable G2U link, there are few obstacles or environment reflection disturbances in the UAV-to-HAP (U2H) link. Therefore, characterized with a wider view, we consider that the U2H link is estimated precisely [33]. Moreover, the OFDMA is adopted in the U2H channel to avoid interferences. Consequently, considering the free space loss and rain attenuation, the maximum achievable rate from UAV n to HAP h is [40], [41], [42]

$$r_n^h = B_h \log_2 \left(1 + \frac{p_h g_n^h L_s L_l}{k_B T_0 B_h} \right), \forall n \in \mathcal{N}, \quad (12)$$

where p_h and g_n^h denote the transmission power and antenna power gain between UAV n and HAP h , respectively. $d_{n,h}$ is the distance between UAV n and HAP h . Moreover, $L_s = (\frac{v_c}{4\pi d_{n,h} f_c})^2$ is the free space path loss. L_l is the total line loss. k_B is the Boltzmann's constant. T_0 is the system noise temperature. B_h denotes the bandwidth. f_c represents the center frequency. v_c is the speed of light. Therefore, the transmission latency and energy consumption for task m from UAV n to HAP h are calculated as:

$$t_{m,n}^h = \frac{\lambda_m^n L_m}{r_n^h}, \forall m \in \mathcal{M}, \forall n \in \mathcal{N}, \quad (13)$$

and

$$E_{m,n}^h = p_h t_{m,n}^h, \forall m \in \mathcal{M}, \forall n \in \mathcal{N}, \quad (14)$$

respectively. Since the backhaul data is much smaller than uplink data, the backhaul delay is ignored [43].

C. Computation Model

1) *UAV-Based Computation Model*: Let f_m represent the CPU frequency allocated to task m . Recall that L_m denotes the data size and c_m is the required number of CPU cycles to compute 1 b data. As a result, the computation latency for processing task m is

$$t_m^{cu} = \frac{c_m L_m}{f_m}, \forall m \in \mathcal{M}. \quad (15)$$

Based on [44], the energy consumption for handling task m is

$$E_m^{cu} = \sum_{n=1}^N (\delta_m^n - \lambda_m^n) \varepsilon_n c_m L_m f_m^2, \forall m \in \mathcal{M}, \quad (16)$$

where ε_n is the effective switched capacitance related to the architecture of MEC servers on UAVs. Note that the CPU frequency of the MEC server is constrained:

$$\sum_{m=1}^M (\delta_m^n - \lambda_m^n) f_m \leq F_{\max}^n, \forall n \in \mathcal{N}, \quad (17)$$

where $F_{\max}^n$ is denoted as the maximum CPU cycle frequency of the UAV n .

2) *HAP-Based Computing Model*: Recall that binary variable λ_m^n represents whether task from GU m is computed at the HAP. Then, in the HAP based computation model, the computing delay and energy consumption for handling task m are

$$t_m^{ch} = \frac{c_m L_m}{f_m}, \forall m \in \mathcal{M}, \quad (18)$$

and

$$E_m^{ch} = \sum_{n=1}^N \lambda_m^n \varepsilon_h f_m^2 c_m L_m, \forall m \in \mathcal{M}, \quad (19)$$

respectively, in which ε_h is the energy consumption coefficient related to the specific chip structure of an MEC server [45]. Let $F_{\max}^h$ denote the maximum computational rate of HAP, the CPU frequency constraint for MEC server of the HAP is

$$\sum_{m=1}^M \sum_{n=1}^N \lambda_m^n f_m \leq F_{\max}^h. \quad (20)$$

Based on the above discussion, the total delay for computing task m is related to the transmission and computation, i.e.,

$$t_m^{total} = t_m^u + \sum_{n=1}^N (\delta_m^n - \lambda_m^n) t_m^{cu} + \sum_{n=1}^N \lambda_m^n t_{m,n}^h + \sum_{n=1}^N \lambda_m^n t_m^{ch}, m \in \mathcal{M}. \quad (21)$$

Moreover, since UAVs are hovering in the air after deployment, the energy consumption for hovering is constant. Therefore, the remaining energy consumption for UAV n is for transmission and computation [27], i.e.,

$$E_n^{total} = \sum_{m=1}^M E_{m,n}^h + \sum_{m=1}^M E_m^{cu}, \forall n \in \mathcal{N}. \quad (22)$$

Besides, since the HAP hovers at the quasi-position, the remaining energy consumption of the HAP is for computation:

$$E_h^{total} = \sum_{m=1}^M E_m^{ch}. \quad (23)$$

D. Problem Formulation

To deal with the potential uncertainties without distribution information, we formulate **P0** with the chance constraints to minimize the total energy cost of the aerial MEC platforms, with restrictions of UAV deployment, task offloading, and resource limitation, i.e.,

$$\mathbf{P0} : \min_{\mathbf{v}, \delta, \lambda, \mathbf{f}} \sum_{n=1}^N E_n^{total} + E_h^{total} \quad (24a)$$

$$\text{s.t. } \Pr \{t_m^{total} \leq T_m^{\max}\} \geq \alpha_m, \forall m \in \mathcal{M}, \quad (24a)$$

$$\lambda_m^n \leq \delta_m^n, \forall m \in \mathcal{M}, n \in \mathcal{N}, \quad (24b)$$

$$E_n^{total} \leq E_n^{\max}, \forall n \in \mathcal{N}, \quad (24c)$$

$$E_h^{total} \leq E_h^{\max}, \quad (24d)$$

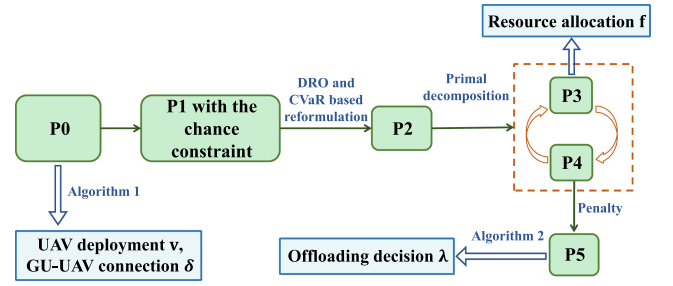


Fig. 2. Overview of the designed algorithms.

$$v_n \in \left\{ (x_n, y_n) \mid X^{\min} \leq x_n \leq X^{\max} \right. \quad (24e)$$

$$\left. Y^{\min} \leq y_n \leq Y^{\max} \right\}, \forall n \in \mathcal{N}, \quad (24f)$$

$$\delta_m^n \in \{0, 1\}, \forall m \in \mathcal{M}, n \in \mathcal{N}, \quad (24g)$$

$$\lambda_m^n \in \{0, 1\}, \forall m \in \mathcal{M}, n \in \mathcal{N}, \quad (24h)$$

$$f_m \geq 0, \forall m \in \mathcal{M}, \quad (24h)$$

$$(4), (6), (17), (20),$$

where in $\mathbf{v} = \{v_n | \forall n\}$ is the UAV deployment positions, $\delta = \{\delta_m^n | \forall m, \forall n\}$ represents GU-UAV connection relationships, $\lambda = \{\lambda_m^n | \forall m, \forall n\}$ denotes the task offloading indicators and $\mathbf{f} = \{f_m | \forall m\}$ is the resource allocation schemes. With respect to the uncertain CSI estimation errors, (24a) is the chance constraint under the uncertainty set $\mathcal{P}$, which indicates that the total latency for processing task m should not be larger than $T_m^{\max}$ with a probability of α_m . Constraint (24b) is the data flow conservation, reflecting the inherent relationship between λ_m^n and δ_m^n . Constraints (24c) and (24d) indicate the total energy consumption of UAV and HAP should not be larger than the maximum capacities $E_n^{\max}$ and $E_h^{\max}$, respectively. (24e) is the constraint for the deployment range of UAVs, wherein $[X^{\min}, X^{\max}]$ and $[Y^{\min}, Y^{\max}]$ are the horizontal and vertical bounds of the area, respectively.

It is observed that **P0** is related with the random parameter Δ_m under uncertainty set $\mathcal{P}$ without distribution information. Besides, **P0** is an MINLP concerning binary variables δ and λ , and continuous variables $\mathbf{f}$ and $\mathbf{v}$, and the time complexity is exponential with the problem scale growing. Therefore, solving **P0** with efficiency is intractable.

IV. ALGORITHM DESIGN

To tackle **P0** efficiently, we divide the process into two phases of UAV deployment and computation offloading. For clarity, the overview of the designed algorithms is illustrated in Fig. 2. As for the UAV deployment, we design a WKD based algorithm in Section IV-A to obtain UAV positions $\mathbf{v}$ and GU-UAV connections δ with a low time complexity. Then, based on the determined pre-deployment of UAVs to handle the CSI estimation error, the CVaR-based mechanism is proposed in Section IV-B. Thus, problem **P1** with the chance constraint (24a) is conservatively approximated and reformulated into **P2** via the DRO

and CVaR based mechanism. In Section IV-C, the reformulated problem **P2** is dealt with by the primal decomposition. **P3** is in the form of SOCP and can be solved via CVX. Furthermore, to effectively obtain the integer offloading strategies of the subproblem **P4**, it is reformulated into **P5**, and the BWOA is designed in Section IV-D.

A. UAV Deployment Optimization

Generally, to reduce the G2U transmission delay, UAVs should be deployed closer to GUs. Moreover, considering the various tasks with different inherent characteristics including data size L_m , computation complexity c_m , and the maximum tolerable delay $T_m^{\max}$, if the UAV is deployed closer to GUs with larger load, the latency is further reduced to better satisfy the QoS. Hence, we design the WKD mechanism to obtain the deployment position $\mathbf{v}$ of UAVs and the GU-UAV connections δ , which highlights the importance of time-sensitive tasks and provides a more practical and efficient solution for the pre-deployment of UAVs. Since the potential uncertainties have a relatively small impact, they are ignored during the pre-deployment operations. The detailed WKD algorithm is provided in Algorithm 1.

First, we select N points in the area as initial positions for UAVs. Then, the distance between UAVs and GUs is obtained according to (1) and all GUs are accordingly assigned to their nearest clusters (line 4). Then, the center point of each cluster is recalculated and the UAV positions are updated as (line 6):

$$v_n = \frac{\sum_{m \in \mathcal{U}_n} \iota_m \mathbf{w}_m}{\sum_{m \in \mathcal{U}_n} \iota_m}, \forall n \in \mathcal{N}, \quad (25)$$

where the weight coefficient ι_m of the task from GU m is obtained via $\iota_m = \varsigma_1 L_m + \varsigma_2 c_m + \frac{1 - \varsigma_1 - \varsigma_2}{T_m^{\max}}$. Specifically, ς_1 and ς_2 are weighted variables for a tradeoff among L_m , c_m and $T_m^{\max}$. $\mathcal{U}_n$ denotes the set of GUs belonging to cluster n . Then, this process is repeated until the result converges and $I_1^{\max}$ is denoted as the number of iterations. After $\mathbf{v}$ is obtained, GUs are accordingly connected with their corresponding UAVs (line 11). During each iteration, the distances between GUs and UAVs are calculated and the positions of UAVs are updated towards convergence [46]. Moreover, since the time complexity of Algorithm 1 is related to the scale of GU M and UAV N , the corresponding time complexity is $\mathcal{O}(MN I_1^{\max})$. During the clustering process, each user is assigned to a cluster and the UAVs are deployed at the weighted cluster centers. Leveraging the proposed WKD algorithm, which is operated based on the distribution of GUs, we obtain the pre-deployment for N UAVs to cover all the clusters as the initial positions for the subsequent operations.

B. CVaR-Based Mechanism for Chance Constraint

From Algorithm 1, we obtain the UAV deployment strategy $\mathbf{v}$ and the GU-UAV connection δ . Thus, the original problem **P0** turns into **P1**, which is only related with task offloading

Algorithm 1: Weighted K-Means Based Multi-UAV Deployment.

Input: Locations of GUs $\mathbf{w}_m$.

1: *Initialization:* Set initial $\mathbf{v}$, $\delta_m^n = 0, \forall m \in \mathcal{M}, \forall n \in \mathcal{N}$.

2: **repeat**

3: **for** $n \in \mathcal{N}$ **do**

4: Calculate the distance $d_{m,n}$ between GU m and UAV n based on (1).

5: Assign GU m to its nearest UAV n^* .

6: Update v_n based on (25).

7: **end for**

8: **until** the result converges.

9: **for** $n \in \mathcal{N}$ **do**

10: **for** $m \in \mathcal{U}_n$ **do**

11: Connect GU m with UAV n , i.e., $\delta_m^n = 1$.

12: **end for**

13: **end for**

Output: UAV deployment location $\mathbf{v}$ and the GU-UAV connection δ .

decision λ and resource allocation $\mathbf{f}$:

$$\begin{aligned} \mathbf{P1} : \quad & \min_{\lambda, \mathbf{f}} \sum_{n=1}^N E_n^{\text{total}} + E_h^{\text{total}} \\ & \text{s.t. } (24a) - (24d), (6), (17), (20), (24g), (24h). \end{aligned} \quad (26)$$

Note that (24a) is the chance constraint, and to deal with it without distribution information and obtain a conservative solution for problem **P1**, we employ DRO to transform (24a) into a distributionally robust chance constraint (DRCC) with uncertainty set $\mathcal{P}$. In detail, let $\inf_{\mathbb{P} \in \mathcal{P}}$ denote the lower bound of possibility for all potential distributions, aiming to seek the solution under the worst case. Then, the chance constraint is reformulated as

$$\inf_{\mathbb{P} \in \mathcal{P}} \Pr_{\mathbb{P}} \{t_m^{\text{total}} \leq T_m^{\max}\} \geq \alpha_m, \forall m \in \mathcal{M}, \quad (27)$$

which is still complicated due to the random parameter Δ_m under the uncertainty set $\mathcal{P}$.

Accordingly, we leverage CVaR mechanism to obtain a conservative estimation for the resource allocation and offloading strategy, which can efficiently improve the reliability while reducing the energy consumption. Generally, CVaR is an indicator to evaluate the risk quantification. It is defined as the conditional expectation value of loss that exceeds a certain probability level under a given probability distribution [47], [48]. The inherent relationship between the loss function $\phi(\xi)$ for random parameter ξ and CVaR under safety factor α is

$$\mathbb{P} \{ \phi(\xi) \leq \mathbb{P} - \text{CVaR}_{\alpha}(\phi(\xi)) \} \geq \alpha. \quad (28)$$

Then, the CVaR constraint in (28) can constitute a conservative approximation for the DRCC, i.e.,

$$\begin{aligned} \sup_{\mathbb{P} \in \mathcal{P}} \mathbb{P} - \text{CVaR}_{\alpha}(\phi(\xi)) &\leq 0, \forall \mathbb{P} \in \mathcal{P} \\ \Leftrightarrow \inf_{\mathbb{P} \in \mathcal{P}} \mathbb{P} \{ \phi(\xi) \leq 0 \} &\geq \alpha, \end{aligned} \quad (29)$$

where $\sup_{\mathbb{P} \in \mathcal{P}}$ is the upper bound under distribution $\mathbb{P}$ [49], [50].

Moreover, referring [51], we obtain Lemma 1.

Lemma 1: For $\Theta \in \mathbb{R}$ and $\theta^0 \in \mathbb{R}$, if the loss function is $\phi(\xi) = \Theta\xi + \theta^0$, the worst-case CVaR $\sup_{\mathbb{P} \in \mathcal{P}} \mathbb{P} - \text{CVaR}_\alpha(\phi(\xi))$ can be derived as a second order cone programming, i.e.,

$$\begin{aligned} \inf_{\beta, e, q, z, s} \quad & \beta + \frac{1}{1-\alpha} (e + s), \\ & e - \theta^0 + \beta + q - \Theta\mu - z > 0, \\ & e \geq 0, z > 0, \\ & \left\| \begin{matrix} q \\ \Theta\sigma \\ z - s \end{matrix} \right\| \leq z + s, \end{aligned} \quad (30)$$

in which β , e , q , z , and s are auxiliary variables. μ and σ are the mean and standard deviation of random parameter ξ , respectively.

Proof: The detailed proof is in Appendix A, available online. $\square$

Hence, the DRCC in (27) can be approximated by a conservative and convex programming problem [52]. Specifically, the complete expression for t_m^{total} is

$$\begin{aligned} t_m^{\text{total}} = & \frac{L_m}{B_u \log_2 \left(1 + \frac{p_u(\bar{g}_m^u + \Delta_m)}{n_0 B_u} \right)} \\ & + \sum_{n=1}^N \frac{\lambda_m^n L_m}{B_h \log_2 \left(1 + \frac{p_h g_h^n L_s L_l}{k_B T_0 B_h} \right)} + \frac{c_m L_m}{f_m}. \end{aligned} \quad (31)$$

Since the estimation error Δ_m is much smaller than the theoretical value of channel gain [53], we adopt the first-order Taylor expansion to approximate the latency t_m^{total} , i.e.,

$$\begin{aligned} t_m^{\text{total}} \approx & \frac{L_m}{B_u \log_2 \left(1 + \frac{p_u \bar{g}_m^u}{n_0 B_u} \right)} + \frac{c_m L_m}{f_m} \\ & + \sum_{n=1}^N \frac{\lambda_m^n L_m}{B_h \log_2 \left(1 + \frac{p_h g_h^n L_s L_l}{k_B T_0 B_h} \right)} \\ & - \frac{L_m \ln 2}{B_u} \frac{p_u \Delta_m}{(n_0 B_u + p_u \bar{g}_m^u) \ln^2 \left(1 + \frac{p_u \bar{g}_m^u}{n_0 B_u} \right)}. \end{aligned} \quad (32)$$

Consequently, the DRCC in (27) is reformulated into

$$\inf_{\mathbb{P} \in \mathcal{P}} \Pr_{\mathbb{P}} \{ \Theta_m \Delta_m + \theta_m^0 \leq 0 \} \geq \alpha_m, \forall m \in \mathcal{M}, \quad (33)$$

where

$$\Theta_m = - \frac{L_m f_m \ln 2}{B_u} \frac{p_u}{(n_0 B_u + p_u \bar{g}_m^u) \ln^2 \left(1 + \frac{p_u \bar{g}_m^u}{n_0 B_u} \right)}, \quad (34)$$

and

$$\theta_m^0 = \frac{L_m f_m}{B_u \log_2 \left(1 + \frac{p_u \bar{g}_m^u}{n_0 B_u} \right)} + c_m L_m$$

$$+ \sum_{n=1}^N \frac{\lambda_m^n L_m f_m}{B_h \log_2 \left(1 + \frac{p_h g_h^n L_s L_l}{k_B T_0 B_h} \right)} - T_m^{\max} f_m. \quad (35)$$

Therefore, according to **Lemma 1**, the DRCC in (27) with random parameter Δ_m is reformulated into an MISOCP, i.e.,

$$\begin{aligned} \inf_{\beta_m, e_m, q_m, z_m, s_m} \quad & \beta_m + \frac{1}{1-\alpha_m} (e_m + s_m) \leq 0, \\ & e_m - \theta_m^0 + \beta_m + q_m - \Theta_m \mu_m - z_m > 0, \\ & e_m \geq 0, z_m > 0, \\ & \left\| \begin{matrix} q_m \\ \Theta_m \sigma_m \\ z_m - s_m \end{matrix} \right\| \leq z_m + s_m, \end{aligned} \quad (36)$$

where β_m , e_m , q_m , z_m , and s_m are all auxiliary variables. Recall that μ_m is the mean value of CSI estimation error Δ_m , and σ_m is the corresponding standard deviation. Thus, via CVaR, **P1** is reformulated as

$$\begin{aligned} \mathbf{P2} : \quad & \min_{\lambda, \mathbf{f}, \beta, \mathbf{e}, \mathbf{q}, \mathbf{z}, \mathbf{s}} \sum_{n=1}^N E_n^{\text{total}} + E_h^{\text{total}} \\ \text{s.t.} \quad & (24\text{b}) - (24\text{d}), (6), (17), (20), (24\text{g}), (24\text{h}), (36), \end{aligned} \quad (37)$$

with the MISOCP constraint in (36) under the worst-case scenario, and a conservative solution can be obtained to enhance the robustness against the fluctuations. β , $\mathbf{e}$, $\mathbf{q}$, $\mathbf{z}$ and $\mathbf{s}$ are the corresponding vectors of β_m , e_m , q_m , z_m and s_m , respectively. However, it is still an MIP problem and complicated to deal with both the binary variables and continuous variables.

C. Primal Decomposition for P2

It is noting that the constraints of problem **P2** can be divided into constraints (24c), (24d), (24h), (17), (20), (36) with respect to variable $\mathbf{f}$, β , $\mathbf{e}$, $\mathbf{q}$, $\mathbf{z}$ and $\mathbf{s}$, as well as constraints (24b), (24d), (24g), (6), (17), (20), (36) related with the binary offloading strategies λ [54]. Specifically, when λ is fixed, the subproblem **P3** related to $\mathbf{f}$, β , $\mathbf{e}$, $\mathbf{q}$, $\mathbf{z}$ and $\mathbf{s}$ is accordingly obtained:

$$\begin{aligned} \mathbf{P3} : \quad & \min_{\mathbf{f}, \beta, \mathbf{e}, \mathbf{q}, \mathbf{z}, \mathbf{s}} \sum_{n=1}^N E_n^{\text{total}} + E_h^{\text{total}} \\ \text{s.t.} \quad & (24\text{c}), (24\text{d}), (24\text{h}), (17), (20), (36). \end{aligned} \quad (38)$$

In the form of SOCP, **P3** can be solved by a standard convex optimization toolkit such as CVX.

With the value of $\mathbf{f}$, β , $\mathbf{e}$, $\mathbf{q}$, $\mathbf{z}$ and $\mathbf{s}$, the offloading decision subproblem **P4** is only related with variable λ :

$$\begin{aligned} \mathbf{P4} : \quad & \min_{\lambda} \sum_{n=1}^N E_n^{\text{total}} + E_h^{\text{total}} \\ \text{s.t.} \quad & (24\text{b}) - (24\text{d}), (24\text{g}), (6), (17), (20), (36). \end{aligned} \quad (39)$$

As a result, **P2** can be handled by iteratively solving **P3** and **P4**. However, **P4** is still intractable to directly solve due to the binary variables.

TABLE I
PARAMETER SETTING

Parameter	Value	Parameter	Value
p_u	$0.5W$	p_h	$2W$
g_0^u	$-50dB$	n_0	$-174dBm/Hz$
B_u	$5MHz$	B_h	$5MHz$
T_0	$1000K$	k_B	$1.38 \times 10^{-23}J/K$
f_c	$2.4GHz$	v_c	$3 \times 10^8m/s$
L_l	$-23dB$	g_m^h	$42dB$
c_m	$300cycles/bit$	T_{max}^m	$20s$
H	10	α_m	95%
F_{max}^n	8×10^9Hz	F_{max}^h	$4 \times 10^{11}Hz$
ε_n	10^{-27}	ε_h	10^{-28}
E_n^{max}	$200J$	E_h^{max}	$20kJ$
μ_m	0	σ_m	$0.1\bar{g}_m^u$
ς_1	0.4	ς_2	0.2

D. BWOA for P4

As for **P4**, the exhaustive search can obtain the optimal solutions. However, as the scale of problem increases, it faces exponential complexity. Hence, we design a meta-heuristic BWOA for efficient solutions, in which each searching agent represents a potential solution for the binary problem. However, since the original BWOA is designed for unconstrained optimization problems, the solution provided by the agent may be not feasible. To address this issue, we adopt the penalty mechanism and reformulate the objective function of **P4** into $\Gamma(\lambda)$, which includes both the objective function as well as the penalty value [55]. Specifically, the agents violating constraints are assigned with a higher fitness value under the influence of penalty factors. As such, the constrained problem is effectively transformed without constraints. Based on above discussions, the fitness function related to λ is defined as:

$$\begin{aligned} \Gamma(\lambda) = & \sum_{n=1}^N E_n^{total} + E_h^{total} \\ & + \sum_{m=1}^M \sum_{n=1}^N \vartheta H_{mn,1}(h_{mn,1}(\lambda)) h_{mn,1}^2(\lambda) \\ & + \sum_{n=1}^N \vartheta H_{n,2}(h_{n,2}(\lambda)) h_{n,2}^2(\lambda) + \vartheta H_3(h_3(\lambda)) h_3^2(\lambda) \\ & + \sum_{n=1}^N \vartheta H_{n,4}(h_{n,4}(\lambda)) h_{n,4}^2(\lambda) + \vartheta H_5(h_5(\lambda)) h_5^2(\lambda) \\ & + \vartheta H_6(h_6(\lambda)) h_6^2(\lambda) + \sum_{m=1}^M \vartheta H_{m,7}(h_{m,7}(\lambda)) h_{m,7}^2(\lambda), \end{aligned} \quad (40)$$

where the penalty factor ϑ is set as 10^5 . $H(\cdot)$ is an index function. $H(h(\lambda)) = 0$ if $h(\lambda) \leq 0$, and otherwise $H(h(\lambda)) = 1$. Hence, by introducing the penalty factor and index function, the solution which violates the constraints leads to an increasing fitness. The penalty factors act as a role to prevent agents from searching infeasible solutions during their explorations and determine

whether the current solution satisfies the corresponding constraints. $h(\lambda)$ is defined based on the constraints of **P4**, i.e.,

$$\begin{cases} h_{mn,1}(\lambda) = \lambda_m^n - \delta_m^n, \\ h_{n,2}(\lambda) = \sum_{m=1}^M \lambda_m^n E_m^h + \sum_{m=1}^M (\delta_m^n - \lambda_m^n) E_m^{cu} - E_n^{max}, \\ h_3(\lambda) = \sum_{n=1}^N \sum_{m=1}^M \lambda_m^n E_m^{ch} - E_h^{max}, \\ h_{n,4}(\lambda) = \sum_{m=1}^M (\delta_m^n - \lambda_m^n) f_m - F_{max}^n, \\ h_5(\lambda) = \sum_{m=1}^M \sum_{n=1}^N \lambda_m^n f_m - F_{max}^h, \\ h_6(\lambda) = \sum_{m=1}^M \sum_{n=1}^N \lambda_m^n - H, \\ h_{m,7}(\lambda) = -e_m + \theta_m^0 - \beta_m - q_m + \Theta_m \mu_m + z_m. \end{cases} \quad (41)$$

Consequently, the objective function of BWOA to tackle **P4** is further transformed as

$$\begin{aligned} \mathbf{P5} : & \min_{\lambda} \Gamma(\lambda) \\ \text{s.t.} & \quad (24g). \end{aligned} \quad (42)$$

Based on the objective function of **P5**, BWOA is further enhanced to search for quasi-optimal solutions, which is a swarm-based technology in light of hunting behaviors of whales [56], [57]. The positions of agents are updated iteratively based on the social behaviors of whales including exploration and exploitation [58]. The quality of each solution is evaluated by calculating the fitness function $\Gamma(\lambda)$ in (40). The position of each agent $X(i_2)$ during iteration i_2 represents a potential solution for λ . Specifically, the procedures of BWOA include encircling prey, spiral updating, and searching for prey, detailed as follows.

1) *Encircling Prey*: For the agent encircling prey, it might update its current position linearly toward the current optimal solution $\vec{X}^*(i_2)$. The position of the agent in the next integration is

$$\vec{X}(i_2 + 1) = \begin{cases} \mathcal{C}(\vec{X}(i_2)), & \text{if } P_{BWOA} < \tau_{ep}, \\ \vec{X}(i_2), & \text{if } P_{BWOA} \geq \tau_{ep}, \end{cases} \quad (43)$$

where τ_{ep} is the step size, which is a possibility to determine whether there is a switch (from 0 to 1 or from 1 to 0) between the current bit value and the value in the next iteration. Specifically,

$$\tau_{ep} = \frac{1}{1 + \exp\left(-10\left(\vec{A} \cdot \vec{D} - 0.5\right)\right)}, \quad (44)$$

where

$$\vec{A} = 2\vec{a} \cdot \vec{r}_1 - \vec{a}, \quad (45)$$

and

$$\vec{C} = 2 \cdot \vec{r}_2, \quad (46)$$

are coefficient factors. The operation $\cdot$ indicates element-wise multiplication [55]. $\vec{a}$ linearly decreases from 2 to 0 during the

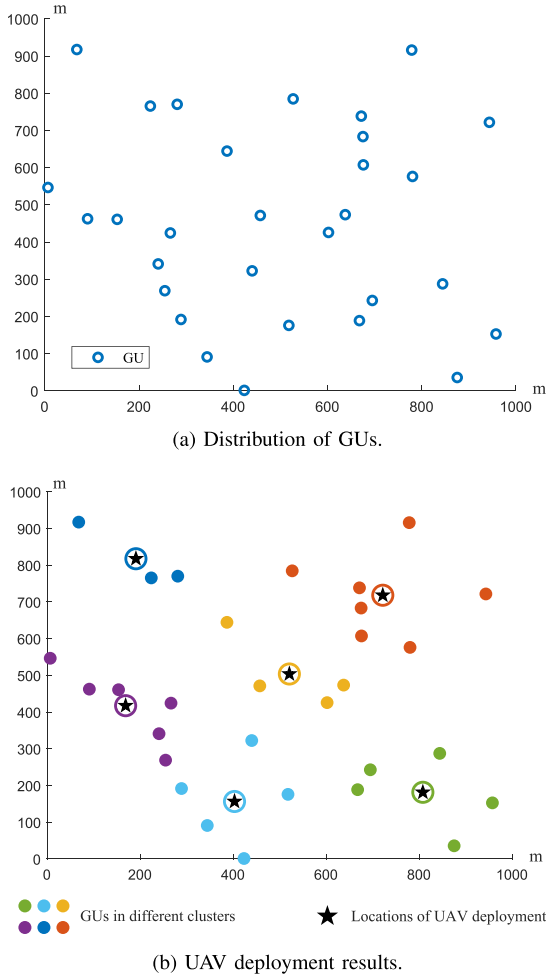


Fig. 3. Cluster results via the WKD algorithm (30 GUs and 6 UAVs in the 1 km × 1 km area).

iteration:

$$a = 2 - i_2 \times \frac{2}{I_2^{\max}}, \quad (47)$$

where i_2 is the index of current iteration, $I_2^{\max}$ is the maximum number of iterations. $\vec{r}_1$ and $\vec{r}_2$ are random vectors within $[0, 1]$. $\mathbb{C}(\cdot)$ represents the complement operation. $P_{BWOA} \in [0, 1]$ is a basis for action selection in the mechanism of encircling prey, and the distance vector $\vec{D}$ is calculated by

$$\vec{D} = |\vec{C} \cdot \vec{X}^*(i_2) - \vec{X}(i_2)|. \quad (48)$$

2) *Spiral Updating*: The agent tends to approach the current optimal individual in either encircling prey or spiral manner. In detail, the spiral updating mechanism for position updating in BWOA is

$$\vec{X}(i_2 + 1) = \begin{cases} \mathbb{C}(\vec{X}(i_2)), & \text{if } P_{BWOA} < \tau_{su}, \\ \vec{X}(i_2), & \text{if } P_{BWOA} \geq \tau_{su}. \end{cases} \quad (49)$$

Moreover, the step size τ_{su} is calculated as

$$\tau_{su} = \frac{1}{1 + \exp\left(-10 \left(\vec{A} \cdot \vec{D}' - 0.5\right)\right)}, \quad (50)$$

Algorithm 2: BWOA for the Offloading Decision.

```

1: Initialization: Set the iteration index  $i_2 = 1$ , maximum
   number of iteration  $I_2^{\max}$  and the agent population  $X_k$ ,
    $k = 1, 2, \dots, K$ .
2: Calculate the fitness value of the search agents and
   obtain the best search agent  $\vec{X}^*(i_2)$ .
3: repeat
4:   for  $k \in \{1, 2, \dots, K\}$  do
5:     Update  $A$ ,  $C$ , and  $a$  according to (45), (46) and
     (47), respectively.
6:   Generate the parameter  $P_{rand} \in [0, 1]$  randomly.
7:   if  $P_{rand} \geq 0.5$  then
8:     Update  $\vec{D}$  by (51) and  $\tau_{su}$  by (50).
9:     Update the position  $\vec{X}(i_2)$  by (49).
10:  else
11:    if  $|A| \geq 1$  then
12:      Select a random agent  $\vec{X}_{rand}$  and update
       $\vec{D}$  based on (54).
13:      Update  $\tau_{sp}$  via (53) and  $\vec{X}(i_2)$  via (52).
14:    else
15:      Update  $\vec{D}$  via (48) and  $\tau_{ep}$  via (44).
16:      Update the positions of agents  $\vec{X}(i_2)$  via
      (43).
17:    end if
18:  end if
19: end for
20: Calculate the fitness value of each agent via (40) and
   obtain the best position  $\vec{X}^*(i_2)$  of the agents.
21: Update the iteration index  $i_2 = i_2 + 1$ .
22: until  $i_2 > I_2^{\max}$ 
Output: The best fitness value  $\Gamma$  and offloading decision  $\lambda$ .

```

where $\vec{A}$ is updated based on (45), and $\vec{D}'$ is updated as

$$\vec{D}' = |\vec{X}^*(i_2) - \vec{X}(i_2)|. \quad (51)$$

3) *Search for Prey*: To achieve more exploration and avoid falling into local optima, some agents conduct random searches instead of updating toward the current optimal. This mechanism is called *search for prey* in BWOA. Since the exploration may bring agents to deviate from the current optima, it enhances the global search capacities of the agents. The updating rule for the positions of agents is

$$\vec{X}(i_2 + 1) = \begin{cases} \mathbb{C}(\vec{X}(i_2)), & \text{if } P_{BWOA} < \tau_{sp}, \\ \vec{X}(i_2), & \text{if } P_{BWOA} \geq \tau_{sp}, \end{cases} \quad (52)$$

where

$$\tau_{sp} = \frac{1}{1 + \exp\left(-10 \left(\vec{A} \cdot \vec{D}'' - 0.5\right)\right)}. \quad (53)$$

Furthermore, $\vec{A}$ is obtained by (45), and $\vec{D}''$ is calculated as

$$\vec{D}'' = |\vec{C} \cdot \vec{X}_{rand}(i_2) - \vec{X}(i_2)|, \quad (54)$$

where $\vec{X}_{rand}$ denotes the position of a random selected agent.

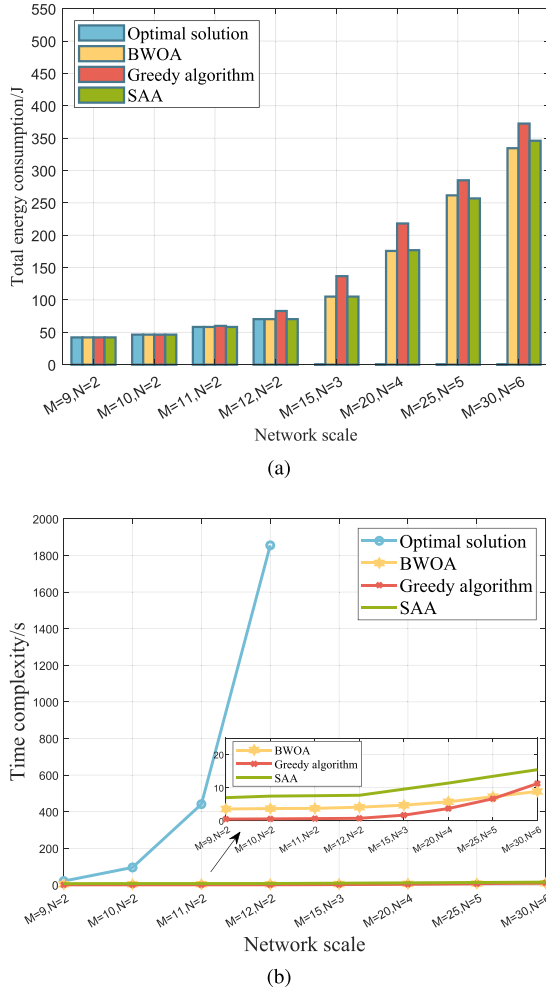


Fig. 4. Performance of different algorithms under different network scales.

4) *Algorithm Design*: The BWOA is provided in Algorithm 2 for solving **P5**. To begin with, K agents are set randomly. Then, the positions of agents are initialized, and the value of fitness function $\Gamma(\lambda)$ for λ can be calculated according to (40). Furthermore, at each iteration, the corresponding parameters a , A and C are updated (line 5). Each agent adopts different strategies based on the probability and updates its position in the next iteration. Specifically, each agent has a possibility of 0.5 to update the parameters and its positions towards optimal in the spiral manner according to (49) (lines 8-9). Otherwise, if the parameter $|A| \geq 1$, the agent is expected to randomly select another agent and search for prey to its direction (lines 12-13). Then, the position in the next iteration is obtained by (52). When $|A| < 1$, the agent encircling prey and linearly approach the individual with best fitness value via (43) (lines 15-16). After all agents updating their positions, their fitness function values are calculated and the position of the current best agent is updated (line 20). The above process is repeated until the result converges [54]. Finally, the offloading decisions concerning λ can be derived. The implementation of Algorithm 2 is mainly related with the number and dimension of agents, i.e., K and MN , respectively. Moreover, the $MN + M + 2N + 3$ constraints of

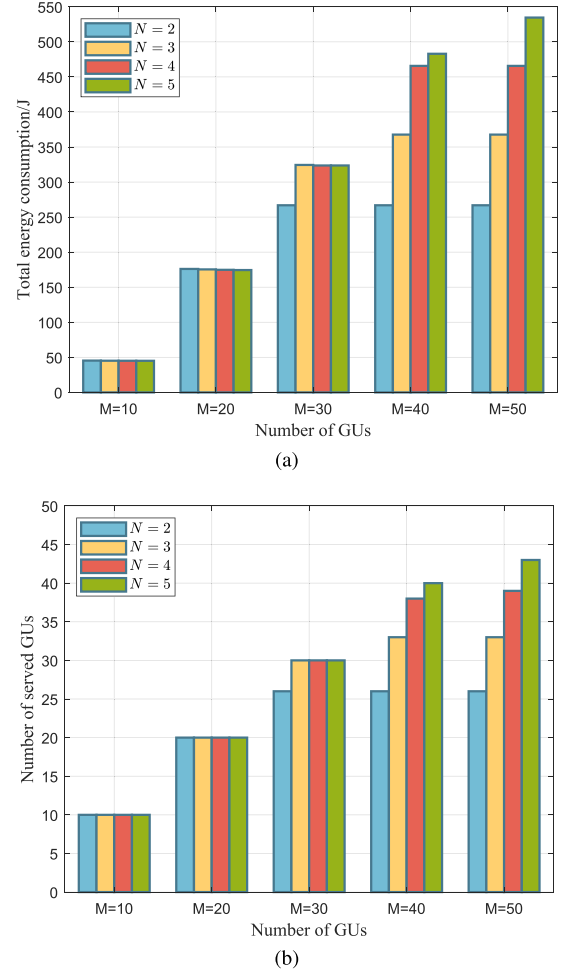


Fig. 5. Energy cost and number of served GUs v.s. the number of GUs with different UAV scale.

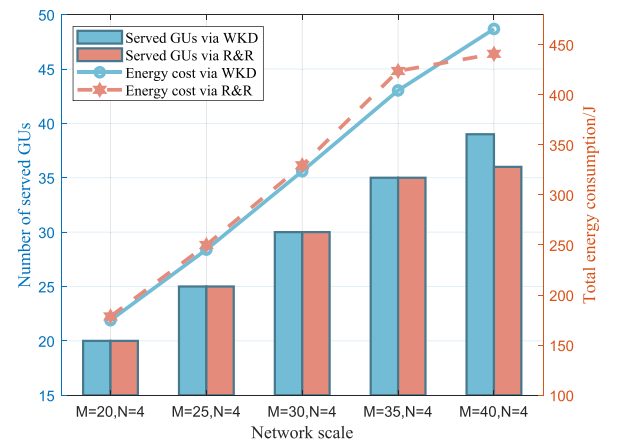


Fig. 6. Performance of the algorithm with deployment optimization.

problem **P4** impact the computational complexity to calculate the index functions. Hence, the complexity of Algorithm 2 is $\mathcal{O}(KMN(MN + M + 2N + 3)I_2^{\max})$.

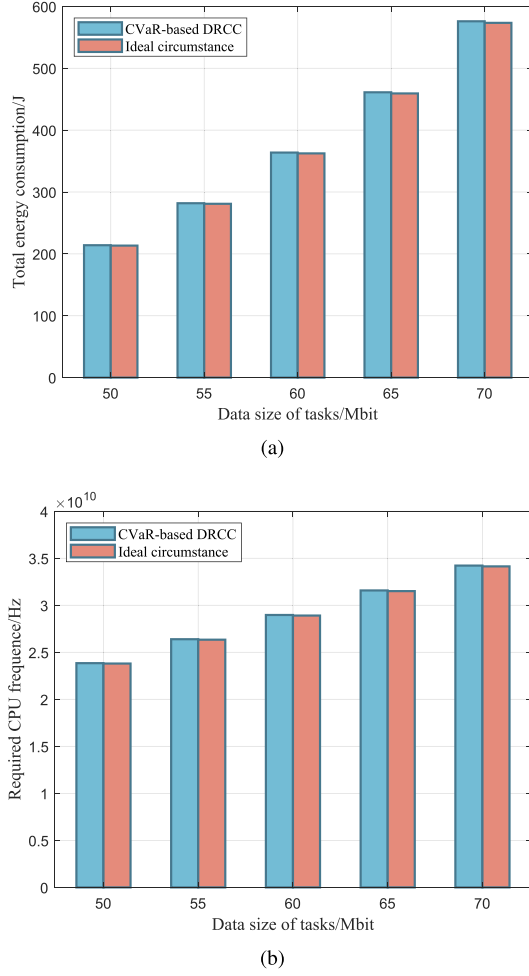


Fig. 7. Performance of the model with CSI estimation errors and ideal conditions.

V. SIMULATION RESULTS

In this section, simulations are conducted to evaluate the proposed algorithms. The GUs are distributed randomly in a 1 km $\times$ 1km area. The altitude of UAVs is 100m and the coordinate of the HAP is $\varpi_h = [500, 500, 2 \times 10^4]$ m. The major parameters are in Table I [32], [33]. The data size of tasks is within [50, 70]Mbits.

The distribution of GUs and UAVs is depicted in Fig. 3(a), including 30 GUs and 6 UAVs. Algorithm 1 is applied to obtain the deployment positions of UAVs, and the clustering results of 30 GUs as well as the positions of 6 UAVs are shown in Fig. 3(b). It is observed that UAVs are deployed at the centers of GU clusters.

To evaluate the effectiveness and efficiency of the proposed BWOA, it is compared with the optimal solution (obtained by exhaustive search), greedy offloading algorithm, as well as the simulated annealing algorithm (SAA), as shown in Fig. 4. Specifically, from Fig. 4(a), it is observed that the optimization results of BWOA are close to optimal and outperform the greedy algorithm and SAA. Meanwhile, Fig. 4(b) provides the corresponding time complexity. Although the optimal solution is obtained by the exhaustive search, the time cost is not acceptable

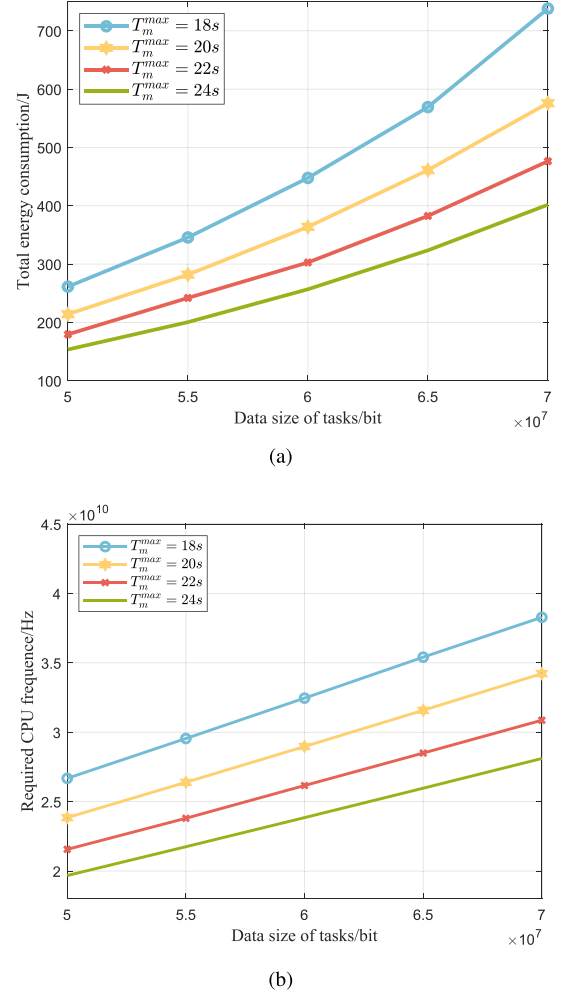


Fig. 8. Impact of tolerable delay on the network performance.

in the large-scale situations. Besides, the time cost of BWOA is lower than the greedy algorithm and SAA with the network scale increasing. Hence, the BWOA shows near optimal performance with lower time complexity.

Fig. 5 depicts the performance of energy cost and the number of served GUs with different number of UAVs. As the number of served GUs increases, more energy is consumed as expected. Moreover, with the same number of served GUs, the energy consumption is almost unchanged by simply increasing the number of UAVs. Besides, note that there exist unserved GUs if the resources are inadequate or the QoS demand of tasks cannot be comprehensively satisfied, indicating that the number of GUs in the network exceeds the network's capacity and the constraints are violated with penalty factors. This is on accounting for the insufficient UAVs which leads to the limited aerial resources and less served GUs. In such a case, the number of served GUs can be improved by deploying more UAVs to alleviate load pressure.

The superiority of proposed WKD based algorithm is verified in Fig. 6. The total energy cost obtained via our proposed algorithms is compared with the results that UAVs are randomly deployed and GUs are randomly connected to the UAVs (R&R). Note that less energy is consumed compared with the baseline

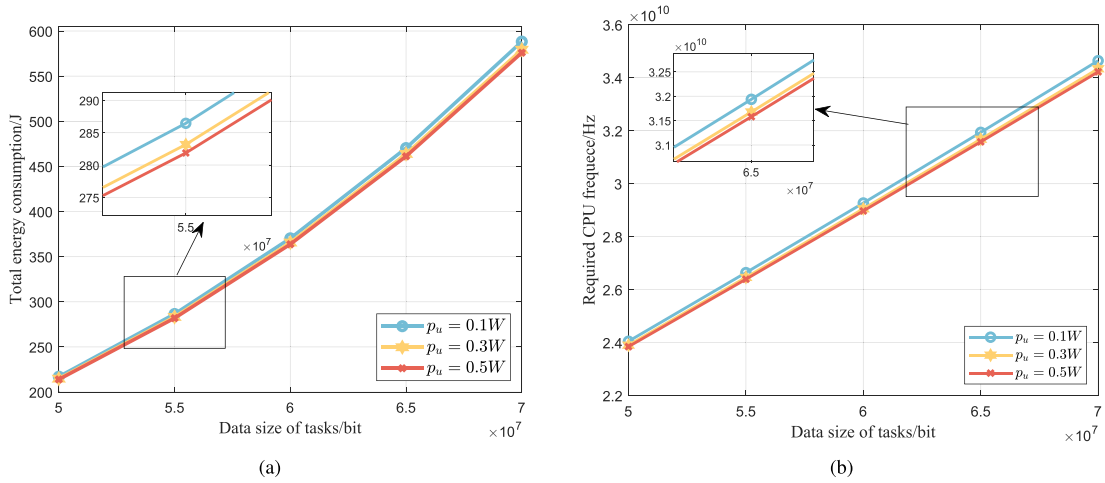


Fig. 9. Impact of transmission power of GUs.

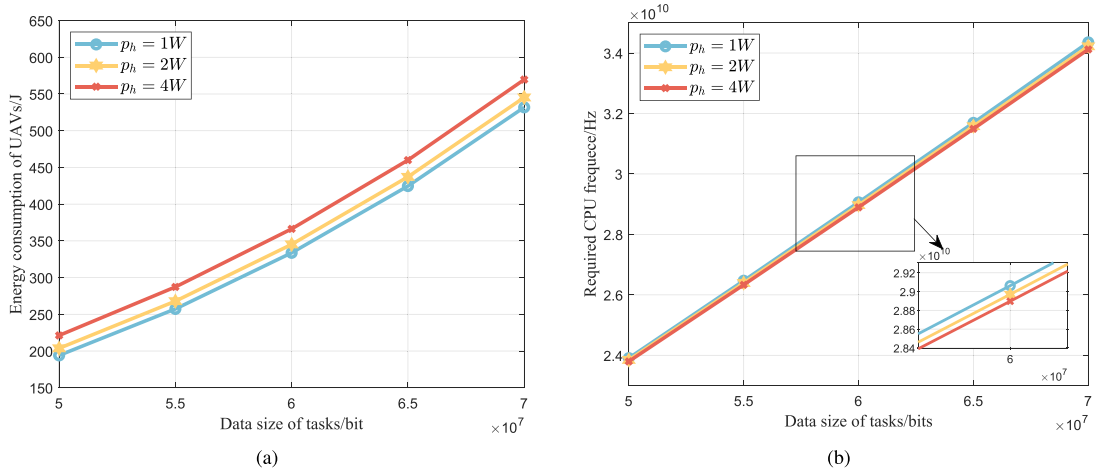


Fig. 10. Impact of transmission power of UAVs.

result with the same served GUs. Moreover, compared with the R&R method, the proposed WKD method can increase the number of GUs that the system can accommodate and make a better use of available resources. Besides, the overloading or under utilization of UAVs can be avoided.

Fig. 7 verifies the robustness of the proposed mechanism (CVaR-based DRCC). As provided in Fig. 7(a), we compare the results optimized via DRCC and CVaR based method with the "ideal circumstance" to evaluate the impact of imperfect CSI, in which the accurate CSI is obtained. When there exist CSI estimation errors, more energy is consumed compared with the ideal circumstance. This is accounted by the fact that the MEC servers allocate more computing resources for the tasks to cope with the impact of environmental disturbances, as shown in Fig. 7(b).

The impacts of the tolerable delay of tasks are shown in Fig. 8. As the maximum tolerable delay increases, the energy consumption decreases, due to the less consumption of computing tasks in Fig. 8(a), since it provides more time for MEC processing and the required CPU frequency is decreased, as shown in Fig. 8(b). Hence, the computation energy is decreased. In addition, Fig. 9

shows the influence of GU transmission power. It is observed that with the increment of transmission power of GUs, both the total energy consumption and the required CPU frequency are decreased. It is explained that the transmission delay is decreased due to the increment of transmission rate, so the time is sufficient for computation.

The effect of transmission power of UAVs on the performance is discussed in Fig. 10. As expected in Fig. 10(a), UAVs consume more energy with the increment of transmission power. Moreover, in Fig. 10(b), more transmission power of UAVs leads to less CPU frequency consumption. It is explained that the transmission rate for G2U link is growing with the increment of transmission power, and thus, there is more remaining time for MEC processing, resulting in less CPU frequency required.

VI. CONCLUSION

In this paper, we proposed a hierarchical aerial MEC model consisting of multiple UAVs and an HAP. Considering the limitation of battery capacities, we jointly optimized the UAV deployment strategies, resources allocation and offloading decisions

to minimize the total energy consumption. Taking into account the imperfect CSI affected by the unpredictable environmental factors, we established an uncertainty set for CSI estimation errors and formulated the problem with the chance constraint. As for the solution, we designed the WKD based algorithm for the deployment of UAVs. Moreover, the chance constraint was transformed into a DRCC, and accordingly approximated into an MISOCPP form under the worst case by employing the CVaR mechanism. Additionally, the MIP problem was further decomposed into two subproblems. To tackle the binary subproblem, BWOA was designed. Finally, we conducted extensive simulations to evaluate the robustness and efficiency. The results showed superiority of the proposed algorithm in the near optimal solution and low time complexity compared with other baseline algorithms. In the future works, the dynamic deployment for UAVs will be further investigated.

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